A IS INTEGRAL OVER THE INVARIANT RING FOR A FINITE LOCALLY FREE GROUPOID SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma lemma-finite-locally-free-disjoint-free (canonical tag 03BI). cited at proof lines 4552-4554. Reduction to constant rank.
- Lemma lemma-determinant-trick (canonical tag 03BH). cited at proof lines 4568-4572. The norm polynomial has coefficients in the invariants.
- Lemma algebra-lemma-charpoly (canonical tag 00DX): Cayley-Hamilton. cited at proof lines 4573-4577. P annihilates t(f).

Source reference. Original equation 03BF: sources/stacks-chow/tags.txt, source lines 4346–4346. Cited at statement lines 4546-4547 fixing C; navigation for the setup.

Source reference. Original lemma 03BI: sources/stacks-chow/tags.txt, source lines 4349–4349. Cited at proof lines 4552-4554 to reduce to constant rank r; load-bearing.

Source reference. Original lemma 03BH: sources/stacks-chow/tags.txt, source lines 4348–4348. Cited at proof lines 4568-4572 to see P(x) = Norm_{s}(t(x - f)) has coefficients in C[x]; load-bearing.

Source reference. Original lemma 00DX: sources/stacks-chow/tags.txt, source lines 514–514. Cited at proof lines 4573-4577 to get P(f) = 0 via faithful flatness of t; load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(A)$ and $R = \mathrm{Spec}(B)$ are affine and $s, t : R \to U$ finite locally free. Let $C \subset A$ be as in (03BF). Then A is integral over C.*

Proof. First, by Lemma 03BI we know that (U, R, s, t, c) is a disjoint union of groupoid schemes (U_r, R_r, s, t, c) such that each $s, t : R_r \to U_r$ has constant rank r. As U is quasi-compact, we have $U_r = \emptyset$ for almost all r. It suffices to prove the lemma for each (U_r, R_r, s, t, c) and hence we may assume that s, t are finite locally free of rank r.

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Assume that s, t are finite locally free of rank r. Let $f \in A$. Consider the element $x - f \in A[x]$, where we think of x as the coordinate on $\mathbf{A}^1$. Since

$$(U \times \mathbf{A}^1, R \times \mathbf{A}^1, s \times \mathrm{id}_{\mathbf{A}^1}, t \times \mathrm{id}_{\mathbf{A}^1}, c \times \mathrm{id}_{\mathbf{A}^1})$$

is also a groupoid scheme with finite source and target, we may apply Lemma 03BH to it and we see that $P(x) = \mathrm{Norm}_{s^\sharp}(t^\sharp(x - f))$ is an element of $C[x]$. Because $s^\sharp : A \to B$ is finite locally free of rank r we see that P is monic of degree r. Moreover $P(f) = 0$ by Cayley-Hamilton (Algebra, Lemma 00DX). More precisely, Cayley-Hamilton says the polynomial $s^\sharp(P) \in B[x]$ annihilates the element $t^\sharp(f)$. Since the coefficients of P are in C we get that $t^\sharp(P(f)) = 0$ in B. Since t is faithfully flat, we get $P(f) = 0$. $\square$